

ZRENJANIN, SerbiaWMO: **131730**Lat: **45.367N**Lon: **20.417E**Elev: **269**StdP: **14.55**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.4	17.5	8.2	8.4	14.5	12.8	10.6	19.2	19.8	32.8	18.0	36.3	3.7	290	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	93.5	71.0	90.5	70.3	87.5	69.2	73.6	87.3	71.9	85.6	70.6	83.8	5.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	110.0	79.1	67.9	103.6	77.0	66.4	98.1	75.5	37.3	87.7	35.8	86.0	34.6	83.9	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.8	14.4	12.2	5.2	97.6	8.7	3.6	-1.1	100.2	-6.3	102.3	-11.2	104.3	-17.5	107.0		
(5)			4.4	76.0	8.4	2.1	-1.6	77.5	-6.5	78.7	-11.2	79.9	-17.2	81.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	33.2	36.8	45.1	55.1	63.6	69.7	72.9	73.2	64.4	54.7	45.4	35.2
	DBStd	16.17	8.74	9.71	8.63	7.41	6.94	6.71	5.96	6.23	7.03	8.12	8.56	8.42
	HDD50	1833	523	375	198	36	1	0	0	0	0	47	190	463
	HDD65	4794	987	789	617	303	112	30	8	8	96	329	590	925
	CDD50	3362	1	6	47	190	422	590	711	718	432	192	50	3
	CDD65	846	0	0	0	7	67	170	254	261	78	9	0	0
	CDH74	8682	0	0	2	81	628	1744	2715	2764	682	66	0	0
	CDH80	3370	0	0	0	11	136	642	1153	1229	196	3	0	0
Wind	WSAvg	5.3	5.8	6.2	6.4	6.0	5.1	4.6	4.4	4.1	4.5	5.1	5.9	5.7
Precipitation	PrecAvg	22.3	1.4	1.5	1.3	1.9	2.3	3.0	2.4	1.9	1.6	1.3	1.8	2.0
	PrecMax	26.8	3.8	3.9	3.9	3.1	4.4	5.5	5.5	7.3	3.3	4.7	3.2	5.0
	PrecMin	15.6	0.1	0.1	0.0	0.7	0.8	1.1	0.5	0.2	0.1	0.0	0.2	0.0
	PrecStd	3.2	1.1	1.1	0.9	0.7	1.1	1.2	1.4	1.6	0.9	1.4	0.7	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.8	63.9	71.4	81.3	87.2	93.8	96.8	98.7	90.9	79.3	69.2	57.2
		MCWB	50.1	51.1	55.2	62.0	66.5	71.5	71.8	71.0	68.2	63.9	58.2	51.8
	2%	DB	51.3	58.0	67.0	75.9	83.6	90.3	93.1	94.1	85.2	75.7	64.4	53.1
		MCWB	47.0	49.2	53.6	59.2	65.4	70.7	71.4	70.5	66.2	61.9	55.6	48.7
	5%	DB	48.0	53.8	62.8	71.9	80.7	87.3	90.0	90.6	81.1	71.9	60.8	49.4
		MCWB	44.5	47.1	51.5	57.3	64.5	69.7	70.7	69.8	64.9	60.2	53.7	45.8
	10%	DB	45.0	49.9	58.5	68.3	77.4	83.8	86.6	87.0	77.6	68.0	57.4	46.3
		MCWB	42.3	44.9	49.3	55.6	63.0	68.2	69.3	68.8	63.6	58.7	51.9	43.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.5	52.6	57.0	63.1	69.2	75.2	75.5	76.0	70.5	65.4	59.8	52.1
		MCDB	55.1	60.9	68.0	76.9	81.7	87.4	90.2	88.4	84.8	76.7	67.3	55.7
	2%	WB	47.3	50.0	54.4	60.6	67.3	72.8	73.6	73.2	68.0	63.3	56.5	48.9
		MCDB	50.8	57.0	64.8	72.9	79.8	85.4	87.3	87.6	81.3	72.9	63.1	52.6
	5%	WB	44.8	47.7	52.4	58.6	65.7	71.0	72.0	71.4	66.1	61.2	54.4	46.1
		MCDB	47.5	53.2	61.3	69.5	77.7	83.3	85.5	86.0	77.6	69.8	60.0	49.0
	10%	WB	42.4	45.4	50.1	56.7	64.1	69.4	70.6	69.9	64.5	59.3	52.1	43.8
		MCDB	45.0	49.6	57.7	66.1	75.2	81.4	83.7	83.4	74.9	67.1	57.0	46.1
Mean Daily Temperature Range	5% DB	MDBR	9.6	13.0	16.9	18.7	19.6	19.6	21.0	22.0	19.1	17.5	12.7	9.3
		MCDBR	13.5	19.3	22.6	24.3	23.9	24.3	25.7	26.8	24.5	22.1	16.8	12.9
		MCWBR	10.2	12.1	12.2	10.9	9.6	8.4	8.5	8.7	9.4	10.8	9.8	9.7
	5% WB	MCDBR	12.9	17.6	20.6	21.6	22.0	21.6	22.8	23.3	21.5	20.4	16.2	12.0
		MCWBR	10.1	11.5	11.6	10.2	9.5	8.5	8.6	8.8	8.7	10.6	9.9	9.4
Clear-Sky Solar Irradiance	taub		0.346	0.372	0.405	0.461	0.447	0.442	0.449	0.456	0.436	0.399	0.364	0.344
	taud		2.334	2.260	2.208	2.059	2.166	2.216	2.190	2.171	2.196	2.293	2.345	2.350
	Ebn at Noon		235	251	259	253	262	264	259	252	246	238	226	222
	Edh at Noon		25	33	39	49	46	44	45	44	39	31	25	23
All-Sky Solar Radiation	RadAvg		379	647	1047	1461	1776	1964	1969	1751	1221	837	501	309
	RadStd		45	87	130	141	161	170	148	164	134	87	41	46